

TASK-01

Brainwave Matrix Solutions

11- Cloud Computing Intern

Deploy a web application in AWS/Kubernetes/GCP/any other
First report submission in 12 Days

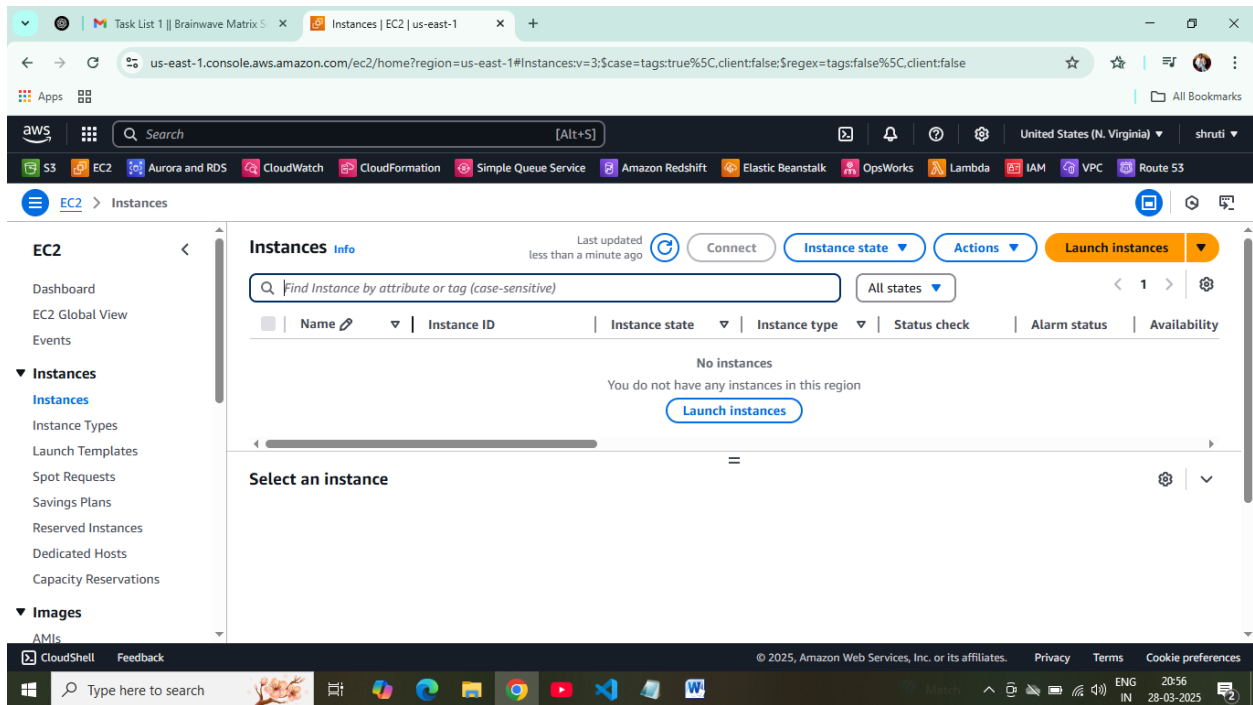
STEP 01: Launch an ec2 instance

- Go to AWS console dashboard.
- Select EC2 from there.
- Click on “launch instances”.
- Give name and tags, select AMI(**ubuntu**).
- Select instance type.
- Create a new key-pair.
- Give proper configurations for the instance.
- Configure security groups
 1. Allow **SSH(port 22)** for remote access.
 2. Allow **HTTP(port 80)** for web traffic.
- Launch the instance.

STEP 02: Connect to the instance & install the web server

- Connect the instance.
- Update the machine and install the web server.
- Commands:
 1. `sudo apt-get update`
 2. `sudo apt install apache2 -y`
 3. `sudo systemctl start apache2`
 4. `sudo systemctl enable apache2`

- Copy the public IP of the instance, paste in browser
- The default page of Apache2 will show.
- Commands:
 1. `cd /var/www/html`
 2. `sudo nano index.html`
 3. Enter the text for your website
 4. Save and exit.
 5. `cat index.html`
 6. refresh the page in the browser, we can now access our application.



Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Recents **Quick Start**

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-04f167a56786e4b09 (64-bit (x86)) / ami-0ae6f07ad3a8ef182 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs [Free tier eligible](#)

Summary

Number of instances [Info](#)

Software Image (AMI)
Amazon Linux 2023 AMI 2023.6.2...[read more](#)
ami-0efc43a4067fe9a3e

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

[Cancel](#) [Launch instance](#) [Preview code](#)

Task List 1 | Brainwave Matrix | Launch an instance | EC2 | us-east-2

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:

Search [Alt+S]

S3 EC2 Aurora and RDS CloudWatch CloudFormation Simple Queue Service Amazon Redshift Elastic Beanstalk OpsWorks Lambda IAM VPC Route 53

EC2 > Instances > Launch an instance

Instance type info Get advice

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour
On-Demand RHEL base pricing: 0.026 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

kp

Create new key pair

Summary

Number of instances info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-04f167a56786e4b09

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Cancel

Launch instance

Preview code

CloudShell Feedback

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Type here to search

28°C Haze 20:58 28-03-2025

Task List 1 | Brainwave Matrix | Launch an instance | EC2 | us-east-2

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LaunchInstances:

Search [Alt+S]

S3 EC2 Aurora and RDS CloudWatch CloudFormation Simple Queue Service Amazon Redshift Elastic Beanstalk OpsWorks Lambda IAM VPC Route 53

EC2 > Instances > Launch an instance

Launching Instance
Creating security group rules 33%

Details

Please wait while we launch your instance.
Do not close your browser while this is loading.

CloudShell Feedback

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Type here to search

28°C Haze 20:58 28-03-2025

The screenshot shows the AWS Management Console for the 'us-east-2' region. The main content area displays the 'Instances (1/1)' page. A table lists the instance 'task-01' with ID 'i-05788c2278393cd97', which is in a 'Running' state. The instance type is 't2.micro' and its status is 'Initializing'. The console also shows a sidebar with navigation options and a top navigation bar with various AWS services.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
task-01	i-05788c2278393cd97	Running	t2.micro	Initializing	View alarms +	us-east-2b

The screenshot shows the 'Connect to instance' page in the AWS Management Console. The page provides options to connect to the instance 'task-01' (ID: i-05788c2278393cd97). The 'Connect using EC2 Instance Connect' option is selected, which uses the public IPv4 address 18.216.61.136. The page also shows the 'Username' field, which is set to 'ubuntu'.

Connect to instance

Connect to your instance i-05788c2278393cd97 (task-01) using any of these options

EC2 Instance Connect | Session Manager | SSH client | EC2 serial console

Instance ID
i-05788c2278393cd97 (task-01)

Connection Type

☒ **Connect using EC2 Instance Connect**
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ **Connect using EC2 Instance Connect Endpoint**
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address
18.216.61.136

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Shreya Sahay

The screenshot displays the AWS CloudShell interface for an EC2 instance. The terminal window shows the execution of `sudo apt-get update`, which lists available updates and provides information about Ubuntu's warranty and update process. Below this, the command `sudo apt install apache2 -y` is partially visible. The interface includes a top navigation bar with AWS services, a search bar, and a bottom taskbar with application icons. The terminal output is as follows:

```
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-20-114:~$ sudo apt-get update

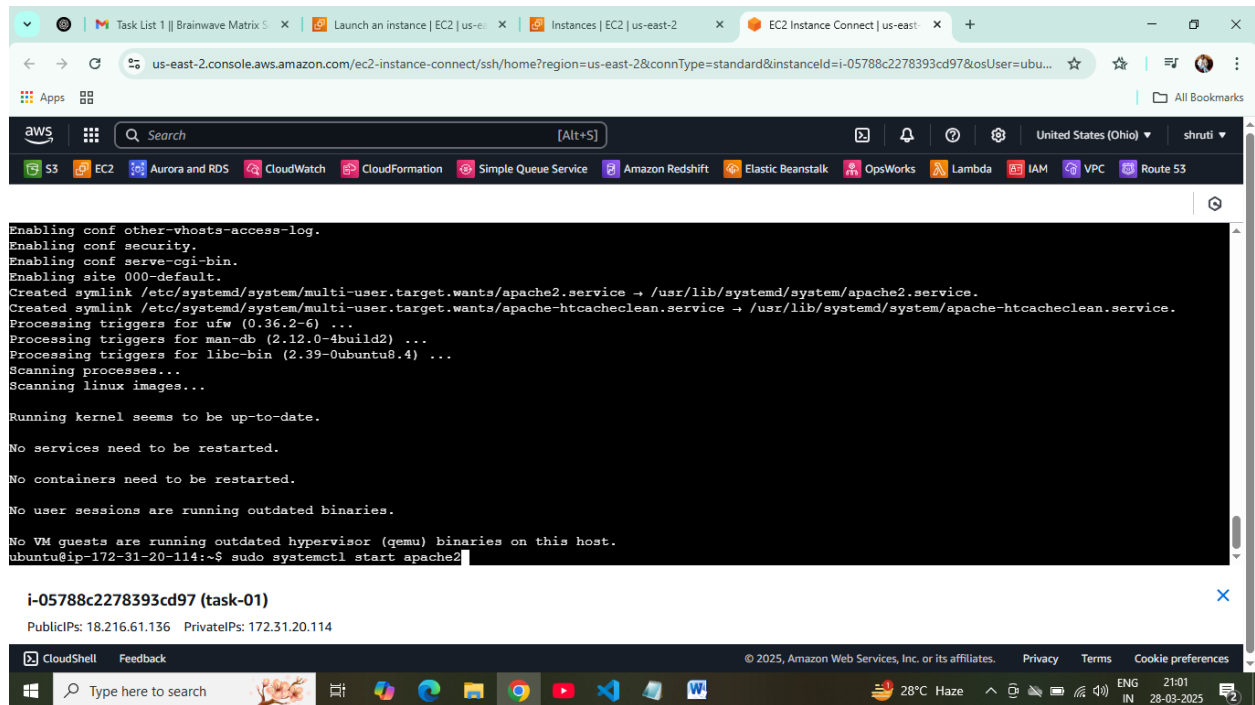
i-05788c2278393cd97 (task-01)
PublicIPs: 18.216.61.136 PrivateIPs: 172.31.20.114

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Type here to search

Get:38 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:39 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:40 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:41 http://us-east-2.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [136 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [8984 B]
Get:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [822 kB]
Get:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [177 kB]
Get:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [51.9 kB]
Get:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Get:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [810 kB]
Get:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [164 kB]
Get:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [460 B]
Get:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.6 kB]
Get:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Get:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Fetched 33.1 MB in 6s (5496 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-20-114:~$ sudo apt install apache2 -y
```

Shreya Sahay



us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-2&connType=standard&instanceId=i-05788c2278393cd97&osUser=ubu...

```
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

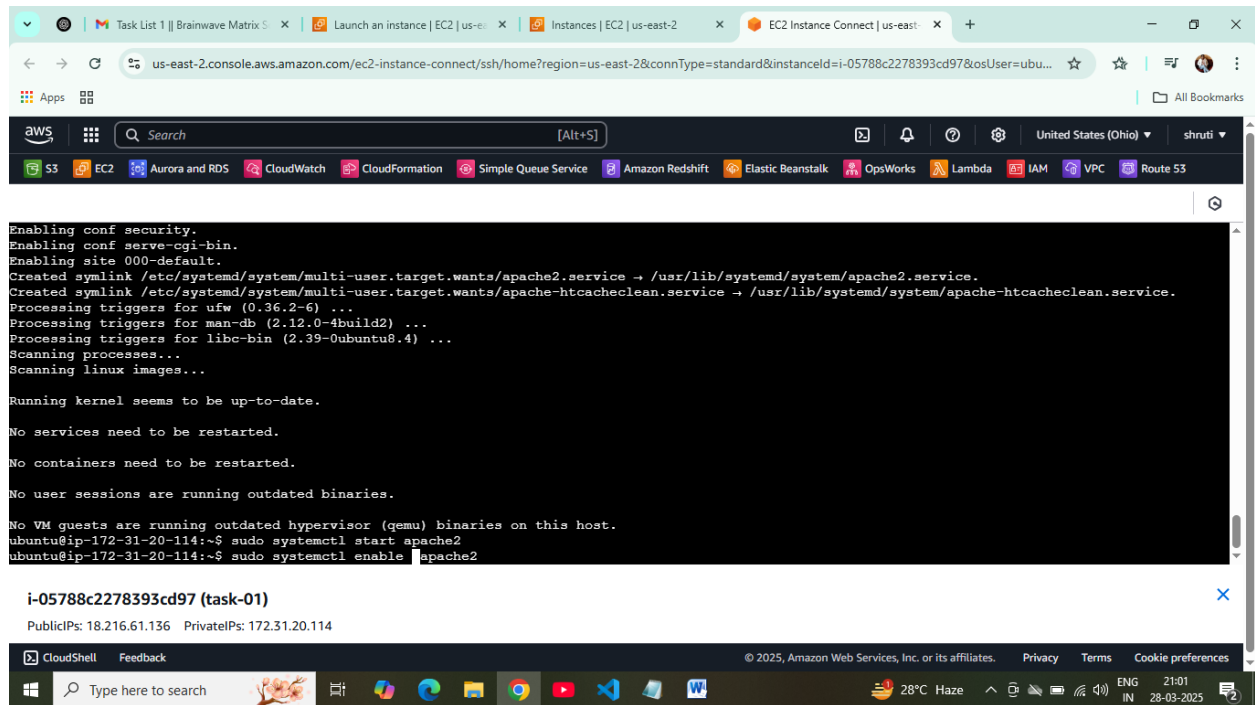
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-114:~$ sudo systemctl start apache2
```

i-05788c2278393cd97 (task-01)

PublicIPs: 18.216.61.136 PrivateIPs: 172.31.20.114

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us-east-2.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=us-east-2&connType=standard&instanceId=i-05788c2278393cd97&osUser=ubu...

```
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-114:~$ sudo systemctl start apache2
ubuntu@ip-172-31-20-114:~$ sudo systemctl enable apache2
```

i-05788c2278393cd97 (task-01)

PublicIPs: 18.216.61.136 PrivateIPs: 172.31.20.114

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Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
 Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
 Processing triggers for ufw (0.36.2-6) ...
 Processing triggers for man-db (2.12.0-4build2) ...
 Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
 Scanning processes...
 Scanning linux images...
 Running kernel seems to be up-to-date.
 No services need to be restarted.
 No containers need to be restarted.
 No user sessions are running outdated binaries.
 No VM guests are running outdated hypervisor (qemu) binaries on this host.
 ubuntu@ip-172-31-20-114:~\$ sudo systemctl start apache2
 ubuntu@ip-172-31-20-114:~\$ sudo systemctl enable apache2
 Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
 Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
 ubuntu@ip-172-31-20-114:~\$

i-05788c2278393cd97 (task-01)
 PublicIPs: [18.216.61.136](#) PrivateIPs: 172.31.20.114

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Apache2 Default Page

Ubuntu

It works!

This is the default welcome page used to test the correct operation of the Apache2 server after installation on Ubuntu systems. It is based on the equivalent page on Debian, from which the Ubuntu Apache packaging is derived. If you can read this page, it means that the Apache HTTP server installed at this site is working properly. You should **replace this file** (located at `/var/www/html/index.html`) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this probably means that the site is currently unavailable due to maintenance. If the problem persists, please contact the site's administrator.

Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default configuration, and split into several files optimized for interaction with Ubuntu tools. The configuration system is **fully documented in `/usr/share/doc/apache2/README.Debian.gz`**. Refer to this for the full documentation. Documentation for the web server itself can be found by accessing the **manual** if the `apache2-doc` package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as follows:

```
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
```


Shreya Sahay

The screenshot displays the AWS CloudShell interface for an EC2 instance. The terminal output shows the following steps:

```
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

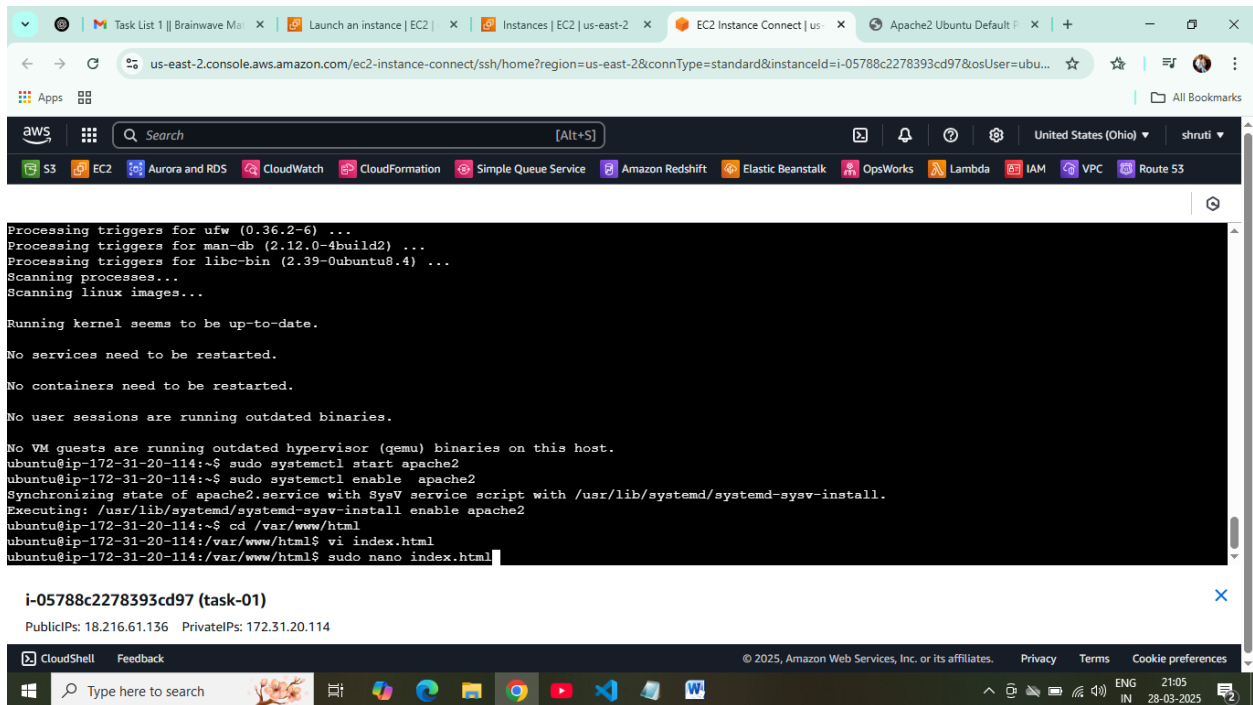
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-114:~$ sudo systemctl start apache2
ubuntu@ip-172-31-20-114:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-20-114:~$ cd /var/www/html
```

The instance ID is **i-05788c2278393cd97 (task-01)**. The public IP is 18.216.61.136 and the private IP is 172.31.20.114.

The terminal also shows the installation of Apache2 and the configuration of the systemd service to start and enable it. The user then navigates to the `/var/www/html` directory and creates a new file named `index.html` using the `vi` editor.

Shreya Sahay



The screenshot shows the AWS Management Console for an EC2 instance named 'i-05788c2278393cd97'. The terminal output displays the following commands and their results:

```
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.4) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

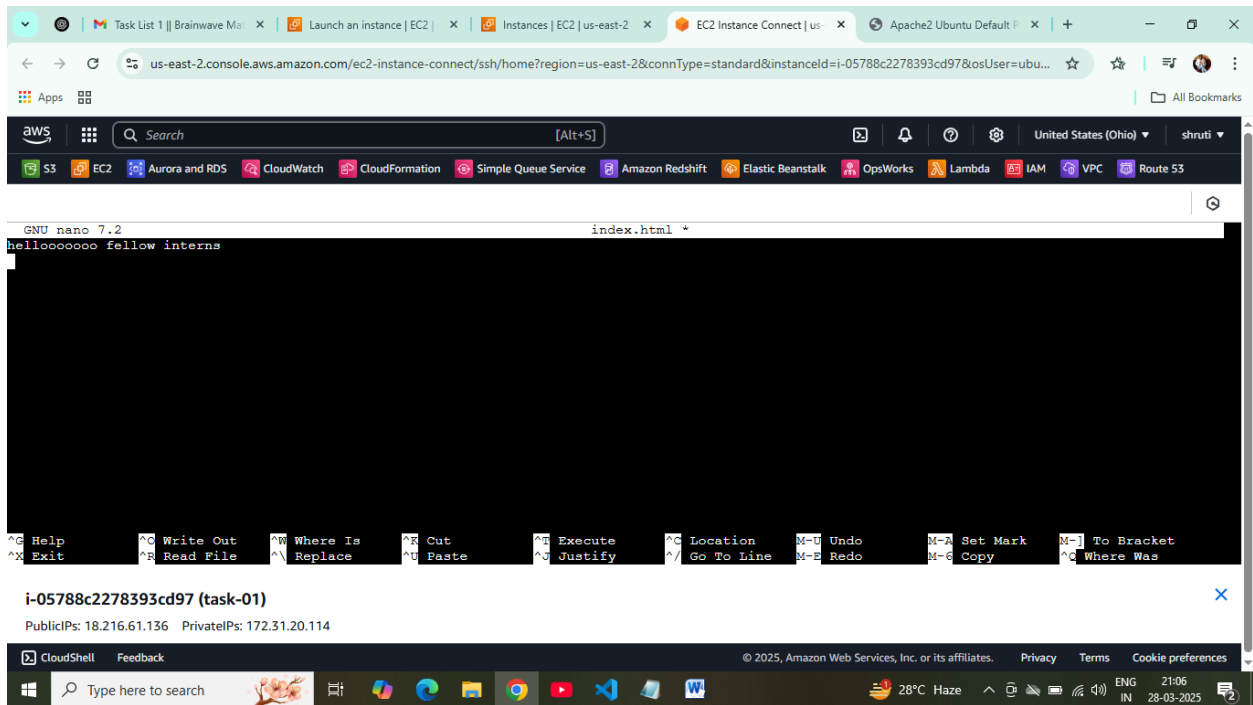
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-114:~$ sudo systemctl start apache2
ubuntu@ip-172-31-20-114:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-20-114:~$ cd /var/www/html
ubuntu@ip-172-31-20-114:/var/www/html$ vi index.html
ubuntu@ip-172-31-20-114:/var/www/html$ sudo nano index.html
```

The instance details show Public IP: 18.216.61.136 and Private IP: 172.31.20.114. The bottom of the screenshot shows a Windows taskbar with various application icons and system tray information.



The screenshot shows the same AWS Management Console terminal window, but now the nano editor is open, editing the file 'index.html'. The content of the file is:

```
GNU nano 7.2 index.html *
helloooooo fellow interns
```

The terminal output shows the user has entered the text 'helloooooo fellow interns' in the nano editor. The bottom of the screenshot shows a Windows taskbar with various application icons and system tray information, including the date and time (21:06, 28-03-2025).

Shreya Sahay

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and various service icons like S3, EC2, Aurora and RDS, CloudWatch, CloudFormation, Simple Queue Service, Amazon Redshift, Elastic Beanstalk, OpsWorks, Lambda, IAM, VPC, and Route 53. The main content area displays the EC2 Instance Connect session for instance **i-05788c2278393cd97 (task-01)**. The session shows a terminal window with the following output:

```
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-20-114:~$ sudo systemctl start apache2
ubuntu@ip-172-31-20-114:~$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
ubuntu@ip-172-31-20-114:~$ cd /var/www/html
ubuntu@ip-172-31-20-114:/var/www/html$ vi index.html
ubuntu@ip-172-31-20-114:/var/www/html$ sudo nano index.html
ubuntu@ip-172-31-20-114:/var/www/html$ cat index.html
helloooooo fellow interns
ubuntu@ip-172-31-20-114:/var/www/html$
```

Below the terminal output, the instance ID **i-05788c2278393cd97 (task-01)** is shown, along with its Public IP **18.216.61.136** and Private IP **172.31.20.114**. The bottom of the console shows the CloudShell interface with a search bar and various application icons.

The screenshot shows a web browser window with the address bar displaying **18.216.61.136**. The browser shows the output of the EC2 Instance Connect session, which is the same text as seen in the terminal window above:

```
helloooooo fellow interns
```

The browser's address bar indicates the connection is "Not secure". The bottom of the browser window shows the Windows taskbar with various application icons and system information like temperature and time.

The screenshot shows the Windows taskbar at the bottom of the screen. It includes the Start button, a search bar with the text "Type here to search", and several application icons including File Explorer, Edge, and various utility apps. The system tray on the right shows the date and time as **28-03-2025** and **21:07**, along with system icons like network and volume.